

A 440-ps 64-bit Adder in 1.5-V/0.18- μ m Partially Depleted SOI Technology

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Abstract—A 64-bit adder in 1.5-V/0.18- μ m partially depleted SOI technology, CMOS8S, and techniques to maintain performance are described. CMOS7S SOI, a 1.8-V/0.22- μ m partially depleted SOI technology, achieves a 28% speed increase over bulk CMOS7S, and CMOS8S SOI delivers an additional 21%. In a 660-MHz CMOS8S SOI processor, the adder compensates for floating body effects in SOI devices which cause history effects, bipolar currents, and lower noise margins on dynamic circuits.

Index Terms—Adders, digital circuits, silicon-on-insulator technology.

I. INTRODUCTION

SILICON-ON-INSULATOR (SOI) technology allows for higher performance than bulk technology. However, the floating body effect in SOI devices poses challenges via history effects, bipolar currents, and lower noise margins on dynamic circuits. This 64-bit adder is used to compute the effective address in a microprocessor [1]. This paper discusses the adder design issues at a high level to give a better understanding of circuit tradeoffs. Adder performance comparison is shown for bulk CMOS7S, a 1.8-V/0.22- μ m technology, CMOS7S SOI, a 1.8-V/0.22- μ m partially depleted SOI technology, and CMOS82 a 1.5-V/0.18- μ m partially depleted SOI technology. Particular emphasis is on the advantages resulting from unique SOI device structures and the techniques for controlling floating body effect in partially depleted devices.

II. HIGH-LEVEL ADDER DESIGN

This adder was originally designed in CMOS7S bulk technology. Then it was converted into CMOS7S SOI. Finally, the adder was converted into CMOS8S2 SOI. This paper focuses on the CMOS8S2 SOI adder. Fig. 1 shows comparisons of the three technologies.

In this high-speed microprocessor [1], cycle time is limited by a number of critical paths, one of which involves a cache memory access by a load or store instruction. For instance, an instruction of the form: lwaux RT, RC, RB (load word Algebraic with update indexed) requires the value in register RC to be added to the value in register RB to produce the effective address (EA). The effective address is used to access the translation lookaside buffer (TLB) and the cache memory as shown in Fig. 2.

Physical, Electrical, and Technology Comparison

	CMOS7S	CMOS7S SOI	CMOS8S2 SOI
Supply Voltage	1.8V	1.8V	1.5V
Poly pitch/2	0.22 μ m	0.22 μ m	0.18 μ m
L_{eff} (NFET)	0.12 μ m	0.12 μ m	0.08 μ m
T_{ox}	3.5nm	3.5nm	2.3nm
Metalization	6 layers Cu	6 layers Cu	7 layers Cu
Contacted M2-M4 pitch	0.81 μ m	0.81 μ m	0.63 μ m
Nominal Temp °C	Y	X	X

Fig. 1. Technology comparison table.

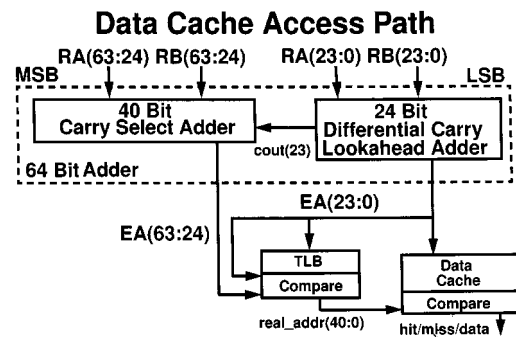


Fig. 2. Data cache access path.

To allow fast TLB and cache memory accesses in this implementation, the adder computes the low-order 24 bits of the effective address faster than the high-order 40 bits. This is accomplished by a high-order 40-bit carry-select adder (CSA) and low-order 24-bit differential carry-lookahead adder (CLA). The carryout of the 24-bit adder, $C_{OUT}(23)$, feeds the 40-bit adder as the carryin.

Fig. 3 shows the internal block diagram for the 24-bit differential CLA. The 24-bit adder is a three-level adder with dynamic buffers inserted as an additional first level. The 24-bit adder has positive logic shown on the left and negative logic on the right. The dynamic buffers receive the RA and RB inputs and generate both true and complement values of the operands. These operands feed the G4 and P4 circuits which generate the 4-bit Generate and Propagate terms for each bit within the adder. The G4_L and P4_L circuits generate the complements. These circuits require large compound domino structures. In compound domino, the output inverter is replaced with a static NOR or NAND gate, which gives both buffering and logic. It has been shown that in SOI, the speedup of NOR and NAND circuits far exceeds the speedup of simple inverters [2]. The next logic levels, shown as G12 and P12, produce the 12-bit Generate and Propagate terms for each bit within the adder, while G12_L and P12_L circuits generate the complements. The HSUM circuit

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64 Bit Adder Block Diagram 24 Bit Carry Lookahead Adder

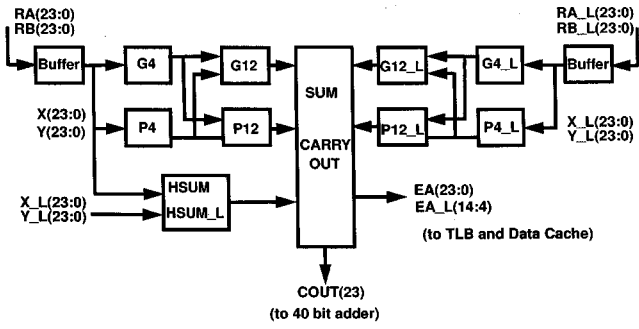


Fig. 3. 24-bit CLA adder.

generates the half sum, an exclusive-OR of the two inputs, while HSUM_L is an exclusive-NOR. The final level of the adder is the SUM-CARRYOUT circuit which generates the final outputs for the 24-bit adder section, EA(23:0) and COUT(23) where $EA_i = HSUM_i \text{ XOR } (G12_{i-1} \text{ or } (P12_{i-1} \text{ and } G12_{i-13}))$ and $COUT(23) = (G12(23) \text{ or } (P12(23) \text{ and } G12(11)))$. EA_L(14:4) is the complement of the effective address EA(14:4) and is sent only to the data cache. Both signals of the data cache address, EA(14:4) and EA_L(14:4) are generated by the 24-bit adder. EA(14:4) and EA_L(14:4) are low during the precharge. During the evaluate phase, each bit's true or complement signal goes high while the other signal remains low. The data cache array begins its access when it detects that all address bits have evaluated with one of their signals going high. This self-timed data cache access solves the chip's worst cycle time limiting path by allowing cycle stealing from the data cache output path to the data cache address path.

Fig. 4 shows the internal block diagram for the 40-bit CSA. The 40-bit CSA section must also make cycle time requirements and is implemented in single-rail dynamic logic to achieve the necessary speed and to keep the area down. A static adder would be too slow and a dynamic dual-rail adder would use more area and power when the speed was not required. The CSA is constructed with 8-bit terms. Each 8-bit term generates sum values assuming a carryin of zero and one. CSA sections often increase in length from least significant bit (LSB) to most significant bit (MSB). In this case, the CSA sections remain at 8 bits because timing is not improved since the late C(23) signal dominates the adder delay. Keeping the CSA sections uniform also makes layout design time shorter. Buses SUM0(63:24) and SUM1(63:24) are the concatenated output buses for the five 8-bit sections. The global carry signals COUT(23,31,39,47,55) select the proper SUM results for their respective 8-bit section. The CSA waits for the COUT(23) from the 24-bit CLA, shown in Figs. 2 and 3. By cascading these two adders, the overall 64-bit carry logic is a maximum of five levels of logic with the majority being single-end domino logic. Thus a balance between speed, power, and area is achieved.

A plot of the 64-bit adder layout shows the division between the 24-bit and 40-bit adder sections in Fig. 5. The LSB is on the left side. The entire 64-bit adder's area is 0.357 mm² in CMOS8S2 SOI technology.

64 Bit Adder Block Diagram 40 Bit Carry Select Section

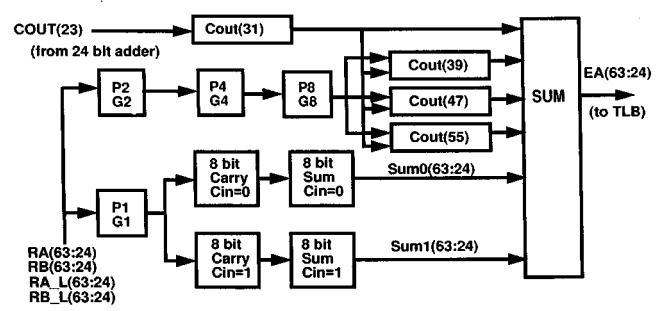


Fig. 4. 40-bit CSA.

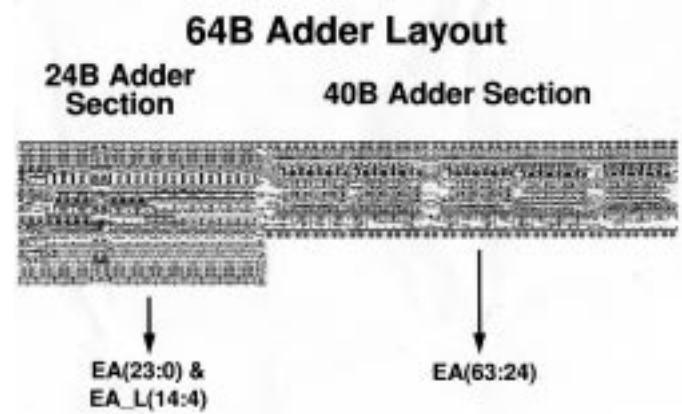


Fig. 5. 64-bit adder layout plot.

III. ADDER PERFORMANCE COMPARISONS

SOI technology provides greater speed than bulk technology via reduced drain-source capacitance, increased current due to lower threshold voltages, and a reduction in the reverse-bias effect on stack transistors. Fig. 6 compares this adder's 7S, 7Ss SOI, 8S2SOI delays from SPICE simulation at nominal conditions shown in Fig. 1. EA(23) and COUT(23) are the MSB and carryout of the 24-bit adder. EA(63) is the MSB of the entire 64-bit adder. For CMOS8S2 SOI, EA(23) is determined in less than 278 ps while the EA(63) is complete in less than 435 ps. There is a 28% improvement moving into the CMOS7S SOI and a 21% improvement going to the CMOS8S2 SOI. The bottom row shows the timing results from a well-known generic circuit Fanout 4 (FO4) which serves as a reference for speed comparison across technologies. The FO4 circuit is an inverter driving four similar size inverters. The values in Fig. 6 are simulated values.

To validate the times in Fig. 6 to hardware, the path from the adder to the data cache is measured in the hardware, as shown in Fig. 7. This path goes through the adder, a latch, and several inverters to move across the chip to the arrays. The timing tools and methodology simulates this path at 681 ps for nominal conditions. The hardware measurement for this path at the same conditions is 675 ps. Fig. 8 shows the best-fit line of data measurements on a number of parts across the spread of the process. The left axis shows the delay for the path from the adder to the

Path Delay Comparisons

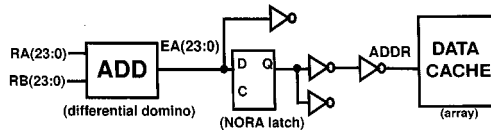
Simulations at nominal conditions

PATH	CMOS7S Bulk	CMOS7S 1st SOI	CMOS8S2 2nd SOI	
	delay	delay	a	b
Cout(23)	426ps	337ps	-21%	283ps
EA(23)	487ps	351ps	-28%	278ps
EA(63)	615ps	489ps	-21%	435ps
FO4	65ps	52ps	-20%	38ps

a. Improvement compared to CMOS7S
b. Improvement compared to CMOS7S SOI

Fig. 6. Adder path delay comparisons.

DCache Address Timing Path



CMOS8S2 SOI: 681ps simulated
CMOS8S2 SOI: 675ps hardware

Fig. 7. Data cache address path timing.

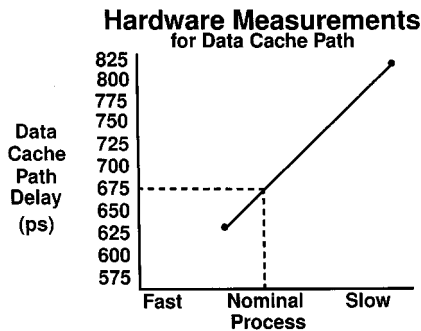


Fig. 8. Hardware measurements' best-fit line.

data cache. The bottom axis shows the process where fast chips are on the left and slow on the right. The dotted line indicates the nominal process condition and points to the hardware speed of 675 ps for the data cache path. Comparing the hardware and simulation delays validates the simulation number as reliable.

Additional delay work to improve the delay on the EA(23) path was done using Logical Effort [9] techniques. After using these techniques, no significant improvements were made. Device sizing before Logical Effort was done by using the driver strength versus load ratio and wire rise/fall time requirements.

IV. FLOATING-BODY EFFECTS IN SOI

SOI technology introduces circuit effects not present in bulk technologies: parasitic bipolar current, lower noise immunity, and history effects. These effects result from allowing the body to hold charge, and thus, vary in potential.

History effect creates a performance range for a FET. The FET performance is determined by the floating body's potential from past switching events. The history effect on delay results in circuit paths becoming faster when they are switched

SOI Bipolar Effect

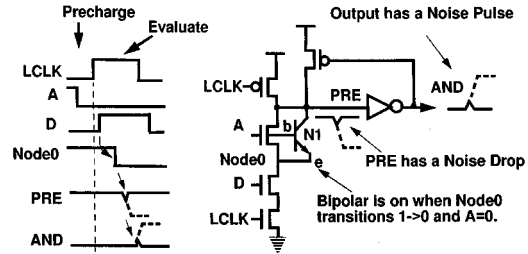


Fig. 9. SOI bipolar effect example on transistor N1.

for the first time after being idle for a long period. To overcome this, timing models assume an "average" history delay. An average history delay is assumed since the delay for an entire latch to latch path determines the chip cycle time. For a path with 10–20 levels of logic, it is very unlikely that all levels will have a worst-case or best-case history effect. The assumption is some levels of logic will have history effects that increase delay and some levels will have history effects that decrease delay, and that these will average over an entire latch-to-latch path. The impact of history effects on overall path delay is 3%–5%. For fast-path or early-mode test, additional margin is given. Fast paths are latch-to-latch paths that have few to no levels of logic between latches. In this case, data intended for the next cycle can arrive before the clock of the destination latch. This means data intended to be latched is lost, and no matter how much the clock cycle is increased, the error will still occur and the chip will not work. For this reason, fast-path or early-mode test must be simulated accurately so that additional margin can be added to insure that a margin exists.

Lower noise immunity from lower threshold voltages requires a stronger noise methodology when designing circuits. A stronger wiring methodology to insure adequate shielding and more circuit simulation for noise analysis are necessary. The guideline for a wire driving an input of a dynamic circuit is that the ratio of the capacitance to other active signals over the total capacitance must be less than 10%: $C_{I-1}/(C_{\text{gnd}} + C_{\text{vdd}} + C_{I-1}) < 0.10$. Guidelines to implement this are as follows.

- 1) Routing a signal wire adjacent to a V_{DD} or GND wire greatly reduces the opportunity for noise due to line-to-line coupling.
- 2) Routing a signal wire between two V_{DD} or GND wires virtually eliminates the opportunity for noise due to line-to-line coupling.
- 3) Avoiding wrong-way wiring on metal layers reduces the opportunity for noise due to line-to-line coupling.
- 4) Separating a sensitive wire from other signal wires by one or more empty wiring channels greatly reduces opportunity for noise due to line-to-line coupling.

Parasitic bipolar current is an unintended current flow from the drain to the source of an otherwise-off FET through a parallel parasitic bipolar transistor. This occurs when the body of a FET, acting as the base of the parasitic n-p-n bipolar junction transistor (BJT), becomes forward biased with respect to the source (in the case of nFET). To illustrate this effect, an example of the

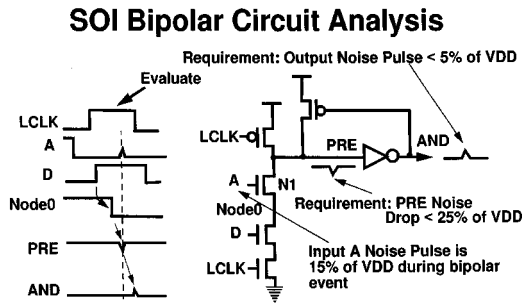


Fig. 10. Adder dynamic circuit noise analysis for SOI.

bipolar effect on the transistor N1 within the dynamic AND circuit is shown in Fig. 9. Notice that the parasitic bipolar transistor is also shown for the N1 transistor. When input A is high during a portion of the precharge cycle, Node0 is charged high. This condition will charge N1's body, which constitutes the base of the parasitic bipolar transistor. If this occurs over many cycles, N1's body voltage will near V_{DD} . At this point, the stage has been set for the bipolar condition to occur. During evaluation, if input A remains low and D goes high, Node0 will fall. This creates a positive voltage across the FET's body and source and the bipolar's base and emitter. The bipolar device is now forward biased and current flows through N1. This bipolar current steals charge from the precharge node PRE, causing a voltage drop, as shown in the timing diagram in Fig. 9. If the bipolar effect is strong, the drop on the precharge node creates a noise pulse on the output, or even flips the circuit's output as shown by the dotted lines in the timing diagram.

Fig. 10 shows the circuit analysis done on each dynamic circuit in the adder. The bipolar event is simulated as shown in Fig. 9. In addition to the bipolar event, a 15% of V_{DD} noise pulse is added to the low input A when the bipolar effect occurs. Both events test if the circuit can handle the worst possible scenario of bipolar current and lower noise immunity. This circuit passes if the precharge node falls less than 25% of V_{DD} and the output pulse rises less than 5% of V_{DD} . More complicated circuits with more than one path from the precharge node to ground are tested with the worst possible scenario. For the case of a dynamic OR circuit with more than four inputs, only four paths from the precharge node to ground were allowed to have bipolar or charge-sharing conditions. The four-path maximum is chosen as a compromise between performance and noise immunity, with the understanding of the low probability of more than four paths having bipolar or charge sharing. Charge sharing is also simulated in a similar fashion as the bipolar event. In most cases, a circuit can only simulate a bipolar event or a charge-sharing event, so these tests are mutually exclusive.

V. TECHNIQUES TO CONTROL BODY EFFECTS IN SOI

Several design techniques improve the noise immunity of dynamic circuits in SOI with minimal impact on delay, as seen by device level simulation. Fig. 11 shows a summary of design techniques used to solve these SOI concerns.

Adding additional conditional feedback fights the precharge node from discharging when test mode is on. Providing more feedback makes the circuit more noise immune. Since most

Design Techniques for Domino Circuits in SOI

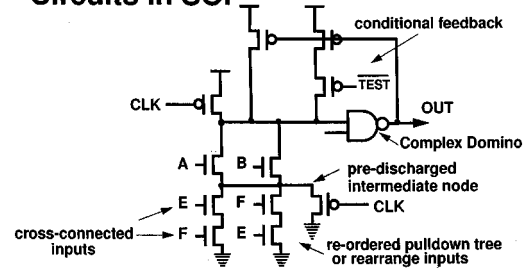


Fig. 11. Summary of SOI domino circuit design techniques.

noise immunity problems occur during the burn-in conditions, conditional feedback is chosen over increasing the original feedback device. Increasing the feedback device requires a permanent tradeoff between performance and noise immunity. If the feedback device is increased to handle the burn-in conditions, the circuit will have slower performance under normal operations when the circuit is more noise immune than necessary. Making the increased feedback conditional permits a choice between noise immunity and speed after the chip is manufactured.

Simple dynamic circuits use an inverter to drive the output from the precharge node. Complex Domino replaces this simple output inverter within the dynamic circuit with a static NAND or NOR circuit. Complex Domino circuits reduce possible noise problems by reducing the number of leaking transistors which are fighting the keeper device. When a NAND circuit is used, the number of paths between the precharge node and ground can be shared across two precharge nodes, which lowers the bipolar effect. In the case of an eight-way OR, two four-way OR precharge nodes can feed a two-way static NAND to lower the bipolar and charge sharing effects, while keeping the same logic function within the circuit. Complex Domino can be used for bulk and SOI technologies to reduce noise problems and increase circuit speed.

Discharging intermediate nodes within the pulldown network prevents the bipolar condition from occurring. Discharging the intermediate nodes avoids charging the FET's body when the intermediate nodes are charged high over a long period of time. Discharging produces charge sharing, but this effect is reduced in SOI technology due to lower drain-source capacitance. Discharging also solves the lower noise immunity problem, since the threshold voltage is higher when the body potential is lower.

Rearranging the inputs can reduce noise problems. A specific example of this is for the dynamic AND circuit. The AND circuits can crossconnect inputs for different fingers to lower the bipolar effect, since only one finger would produce the bipolar effect.

Reordering the pulldown tree can also reduce noise problems. Rearranging the pulldown network can reduce the number of nFETs connected to the precharge node and reduce the bipolar effect by reducing the number of leaking transistors that fight the keeper device.

Another fix for SOI bipolar leakage current is possible when the pulldown network of a dynamic circuit has a dominant input [4]. An input is dominant when input = 0 forces the output = 0. For example, the circuit shown in Fig. 12, $\text{OUT} = A(B + C)$, has A as a dominant input. When a dominant input feeds an

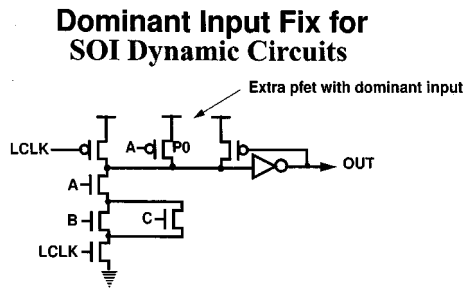


Fig. 12. Dominant input circuit example in SOI.

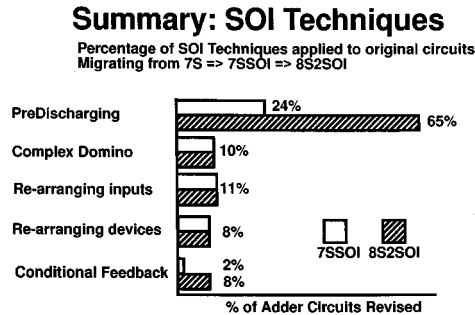


Fig. 13. SOI techniques usage summary.

nFET in the pulldown network of a dynamic circuit, the input can also control an extra precharge transistor, P0, whose drain is attached to the precharge node and source to V_{DD} . This additional pFET, P0, makes the dynamic circuit more immune to SOI bipolar effects by supplying current during conditions where bipolar leakage can occur. This concept does not affect performance since P0 is off when A is high and the precharge node is pulled low by B or C without fighting P0. Since P0 is small, the additional gate, source, and drain capacitances are negligible.

The bar graph in Fig. 13 shows how often these SOI techniques are used within the adder. It shows what percentage of the adder circuits used the various SOI techniques for the two SOI technologies. The white bars show the percentages for the CMOS7S SOI adder and the hashed bars show the percentages for the CMOS8S2 SOI adder. Predischarging was used in 24% of the CMOS7S SOI adder dynamic circuits and 65% in CMOS8S2 SOI. In both SOI technologies, predischarging nodes within the pulldown network is used most often. The increased usage of predischarging when going from CMOS7S SOI to CMOS8S2 SOI (24% to 65%) is due to scaling effects of the technology.

VI. CIRCUIT EXAMPLES

The first circuit example, Fig. 14, is the dynamic buffer that is inserted into the SOI adder as the first logic level. The inputs are from latches, so a foot device is required to insure precharge occurs. A foot device is an nFET in the pulldown network that is gated by the local clock. To prevent the bipolar effect in the circuit, the two inputs are flipped in the pulldown network. The latch input, A , gates the bottom nFET and the local clock gates the top nFET. Since the input A comes from a latch, it must be stable before the local clock rises. Connecting the later arriving clock signal to the top nfet makes the circuit faster. With the

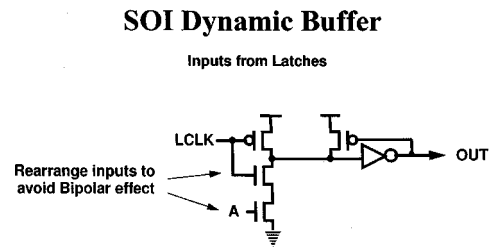


Fig. 14. Dynamic buffer in SOI.

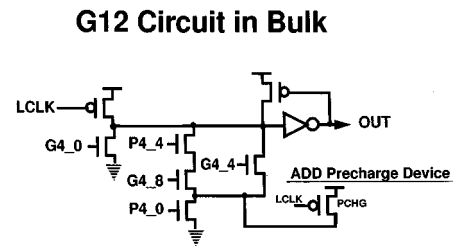


Fig. 15. 12-bit generate circuit in bulk.

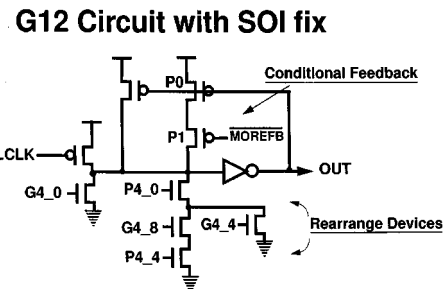


Fig. 16. 12-bit generate circuit in SOI.

clock gating the bottom nFET under CMOS8S2 SOI burn-in conditions, the bipolar effect drops the precharge node 14% of V_{DD} . By switching the inputs, the bipolar effect is eliminated and charge sharing drops the precharge node 9%, so more noise immunity is created.

The 12-bit generate circuit (G12) in the bulk technology is shown in Fig. 15. G12 requires only one noise fix in the bulk technology, adding a precharge device, PCHG, to avoid a charge-sharing condition. Notice the pulldown network has three nFETs in the pulldown network connected to the precharge node. The 12-bit generate circuit in SOI has three changes, as shown in Fig. 16. The pulldown network is rearranged to remove one transistor connection to the precharge node. This eliminates one path of bipolar effects. Additional conditional feedback improves noise immunity when the precharge node begins to fall. MOREFB stands for more feedback and is high during stress testing or burn-in conditions and low during normal operation when higher performance is required. The third change is removing the precharge device. The bulk 12-bit generate circuit under SOI burn-in conditions has a 24% of V_{DD} drop on the precharge node from bipolar effects. The SOI version has a 14% drop on the precharge node for bipolar and a 16% drop for charge sharing. The bipolar and charge-sharing cases are mutually exclusive cases, since the circuit can only have charge-sharing or bipolar effects, not both simultaneously.

Bulk Sum Circuit, Dual Rail Output

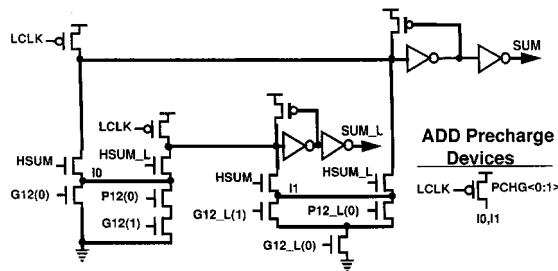


Fig. 17. Dual-rail sum circuit in bulk.

SOI Sum Circuit, Dual Rail Output

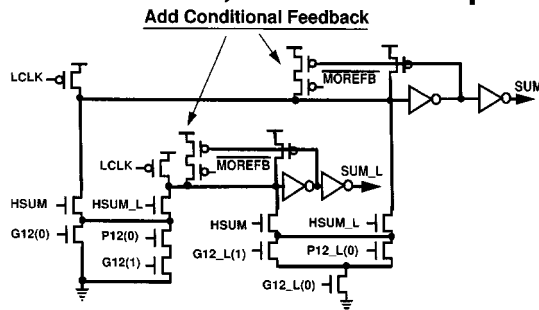


Fig. 18. Dual-rail sum circuit in SOI.

The dual-rail sum circuit sends both the true and complement signals of the effective address to the data cache. Fig. 17 shows the circuit in bulk technology. The circuit requires precharge devices, PCHG(0:1), for charge-sharing noise immunity. The SOI version, shown in Fig. 18, has the precharge devices removed and conditional feedback added to both outputs. The additional inverters on the output are for repowering to drive a long wire to the data cache and add more noise immunity. The bulk sum circuit under SOI burn-in conditions has bipolar noise effects that dominate over charge-sharing noise. This circuit has a $>25\%$ of V_{DD} drop on the precharge node from bipolar effects and an 12% of V_{DD} drop from charge sharing. The SOI version has a 19% drop on the precharge node for bipolar and a 14% drop for charge sharing.

The 4-bit generate circuit, G4, is shown in bulk technology in Fig. 19. This circuit uses the complex domino NOR structure for greater performance. The NDIS transistor is a discharge transistor that prevents a charge-sharing event between node INT_OUT and the output G4_OUT. The precharge devices, PCHG(0:4), are added to precharge internal nodes and prevent charge sharing. This circuit is the first level of the bulk adder. It receives its inputs from latches, so foot devices are required in the pulldown network. This same circuit in SOI is the second level of logic within the adder, shown in Fig. 20. The first level is the dynamic buffers. This allows the removal of the foot devices. Removing the foot devices eliminates one of the bipolar conditions and increases the speed of this circuit to offset the additional delay from the dynamic buffers. The path delay did not change the G4 gained noise immunity.

Although removing the foot device helps, it does not fix the entire noise problem. SOI bipolar leakage from the precharge

G4 Circuit in Bulk

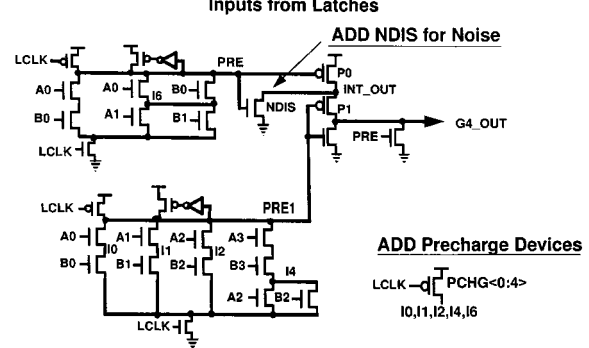


Fig. 19. Four-bit generate circuit in bulk.

G4 Circuit with SOI Fixes

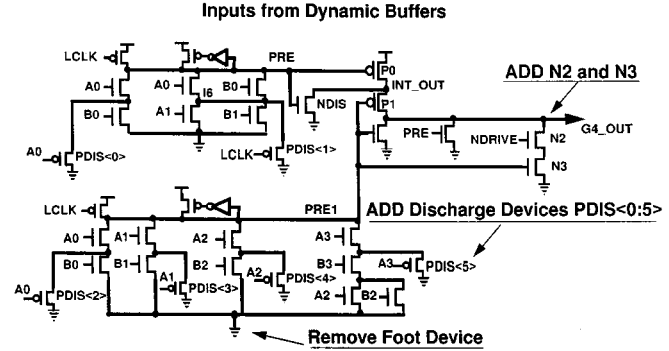


Fig. 20. Four-bit generate circuit in SOI.

node to intermediate nodes can still occur. Therefore, internal nodes within the pulldown network of G4 are discharged during the precharge phase to reduce SOI bipolar effects. Discharging these internal nodes guarantees charge sharing within the G4 circuit. However, this effect is reduced in SOI technology due to lower drain-source capacitance. In Fig. 20, six discharge devices, PDIS(0:5), are added for the G4 circuit. Note that circuit data inputs (A0, A1, A2, and A3) drive these discharge devices, except in the case of PDIS(1) which must use the local CLK input to avoid a dc path from V_{DD} to GND. This causes minimal effects to performance while preventing SOI bipolar effects.

G4 is a compound dynamic domino circuit using a static NOR. The NOR circuit requires P0 and P1 to be large since they are in series and part of the evaluate path making the beta ratio of the NOR high. A problem occurs when the node PRE discharges and the node PRE1 falls slightly from noise effects. In this state, an output noise pulse can be generated due to the high beta ratio of the NOR. To solve this problem, two additional nFETs, N2 and N3, are added to the static NOR [3]. These nFETs alter the beta ratio of the NOR, allowing more noise immunity for PRE1. The intent is to have NDRIVE high during burn-in or stress testing and low during normal operation. NDRIVE is the complement of the MOREFB signal. NDRIVE and MOREFB are two separate signals to provide flexibility in controlling the more feedback and more drive functions independently. NDIS also performs this ratio balancing but for noise immunity on the PRE node. The bulk G4 circuit under SOI burn-in conditions has a

>25% of V_{DD} drop on the precharge node from bipolar effects. The SOI version has a 21% drop on the precharge node for bipolar and a 16% drop for charge sharing.

VII. CONCLUSION

This paper shows that partially depleted SOI technology gives higher performance than bulk technologies. Performance comparisons indicate a 28% speed increase from CMOS7S bulk to CMOS7S SOI. There is a 21% speed increase when moving from CMOS7S SOI to CMOS8S2 SOI. Overall, the adder speedup is over 43%, giving a complete 24-bit add in less than 280 ps. The less critical 64-bit path speedup is over 30% with path delay less than 440 ps. Challenges in SOI dynamic circuits from the floating-body effect are:

- bipolar current;
- lower noise margin;
- history effect.

There are six circuit techniques used to address the bipolar current and lower noise margin challenges.

- 1) Add extra feedback that is conditional with a test signal to add more noise immunity, instead of making the feedback device bigger.
- 2) Predischarge internal nodes to eliminate bipolar events and prevent lower noise margin.
- 3) Use complex domino to reduce the load on precharge nodes.
- 4) Rearrange inputs in the pulldown network to avoid bipolar events.
- 5) Rearrange the pulldown network to avoid bipolar events.
- 6) Add extra precharge device that is gated by dominate input.

These concepts are used within 65% of the dynamic circuits. Predischarging is used most often in the CMOS7S SOI and CMOS8S2 SOI adders. History effect concerns are handled by using timing models that assume an average history delay.

This paper shows it is possible and desirable to design processors or other VLSI designs in partially depleted SOI technology.

REFERENCES

- [1] T. Buchholtz, "A 0.18- μm 1.5-V SOI 660-MHz 64-bit PowerPC™ microprocessor with copper interconnects," in *ISSCC Dig. Tech. Papers*, Feb 2000, pp. 88–89.
- [2] D. Allen, "A 0.2 μm 1.8-V SOI 550-MHz 64-bit PowerPC™ microprocessor with copper interconnects," *IEEE J. Solid-State Circuits*, vol. 34, pp. 1430–1435, Nov. 1999.

- [3] A. Davies and D. Stasiak, *Method and Apparatus to Prevent Output Noise in NOR Complex Domino Dynamic Circuits*. New York: IBM Research Disclosure, 1999.
- [4] —, "Method and apparatus for using circuit input to reduce bipolar current effects in dynamic circuits designed in silicon-on-insulator," patent pending, 1999.
- [5] G. Shahidi, "Partially depleted SOI technology for digital logic," in *ISSCC Dig. Tech. Papers*, Feb 1999, pp. 426–427.
- [6] H. Ngo, "High-speed binary adder having inclusive propagates and generates," 0116-AD-AT9-97-509 IBM, patent pending.
- [7] N. Weste and K. Eshraghian, *Principles of CMOS VLSI Design: A Systems Perspective*. Reading, MA: Addison-Wesley, 1985, pp. 310–338.
- [8] D. Allen and D. Stasiak, "Method and apparatus to control noise in a dynamic circuit," U.S. Patent 6 002 292, Dec. 14, 1999.
- [9] I. Sutherland, B. Sproull, and D. Harris, *Logical Effort Designing Fast CMOS Circuits*. San Mateo, CA: Morgan Kaufmann, 1999.



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